Games, processes and constructions

Kevin Hu
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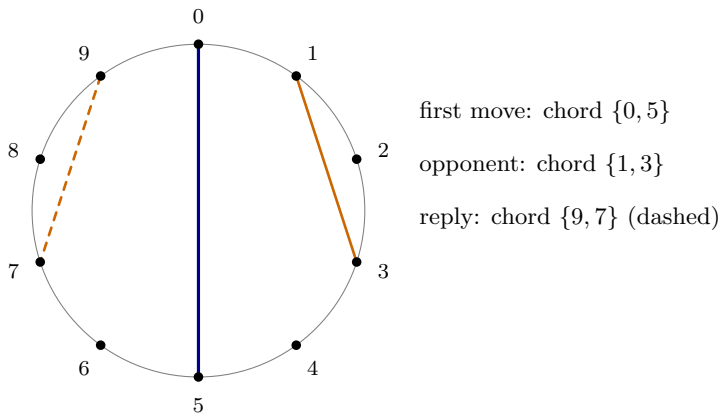
L5-CO2 Level 5 Advanced · Combinatorics

Winning and losing positions, symmetry, pairing and strategy stealing, termination of processes, and constructions that meet a bound, including a first look at the averaging (probabilistic) method.
By the end you can classify positions of a game and prove a strategy correct, including pairing, symmetry and strategy-stealing arguments; prove that a process terminates and describe its final states; prove existence by averaging over random choices and match the bound with a construction.
Prerequisites: Invariants, monovariants and extremal arguments (L4-CO1), Recursion, bijections and double counting (L4-CO2). **Contests:** BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

There are $2n$ points on a circle. Two players take turns to draw a chord joining two of the points, where neither point may already be used and the chord may not meet any chord already drawn. The player who cannot move loses. Who wins?

The first player. Number the points $0, 1, \ldots, 2n-1$ in order and draw the chord from 0 to n. The chord leaves $n-1$ points on each side. Every later chord lies on one side, since a chord with ends on both sides would cross the first one. Whenever the opponent draws a chord, reply with its mirror image in the line of the first chord. By symmetry the mirror chord uses two free points. This chord lies on the other side, so the chord meets neither the opponent's chord nor any chord on that side, and meets no chord on its own side, because the position before the opponent's move was symmetric. So the first player always has a reply, and since the game is finite, the opponent eventually cannot move.



This unit is about problems of three kinds, namely games (who wins, and how), processes (whether the process stops and where the process ends), and constructions (build an object

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that meets a bound). In each you must prove that something exists (a strategy, a final state or an example), and the proof is usually an explicit description followed by a check. I assume invariants and monovariants from L4-CO1 and the graph theory of L5-CO1.

1.1 Winning and losing positions

Consider a game between two players who move alternately, where the available moves depend only on the position (not on who is moving), each position allows finitely many moves, every play ends after finitely many moves, and the player who cannot move loses. Call a position a *P-position* if the player who has just moved (the previous player) can force a win, and an *N-position* if the player about to move (the next player) can.

Theorem 1.
Every position is either a P-position or an N-position. A position is a P-position if and only if every move from that position leads to an N-position. In particular a position with no moves is a P-position. A position is an N-position if and only if some move leads to a P-position.

Proof. Induct on the length of the longest play from the position, which is finite by assumption. If no move is available, the next player has lost. Otherwise every position reachable in one move is P or N by induction. If one of them is P, the next player moves there and from then on is the previous player in a P-position, so wins. If all are N, whatever the next player does, the opponent becomes the next player in an N-position and wins. ■

So to solve such a game, compute the P-positions for small cases, guess the pattern, and then prove the guess by checking the two properties. No move from a claimed P-position reaches another claimed P-position, and from every other position some move reaches one. You do not need to describe the whole game tree.

Example 1.
A pile contains 2026 stones. Two players alternately remove stones. In each move a player must remove at least one stone and at most half of the stones currently in the pile. The player who cannot move loses. Who wins, and how?

Solution.
A pile of 1 stone allows no move, so 1 is a P-position. From 2 the only move is to 1, so 2 is N. From 3 the only move is to 2, so 3 is P. Continuing, the P-positions up to 20 are $1, 3, 7, 15$.
Guess. The P-positions are the numbers $2^k - 1$ with $k \geq 1$.
Check the two properties. From $n = 2^k - 1$ a player removes at most $2^{k-1} - 1$ stones (half of n, rounded down), leaving between 2^{k-1} and $2^k - 2$ stones, which is not of the form $2^j - 1$. From any other $n \geq 2$, choose k with $2^k - 1 < n < 2^{k+1} - 1$ and remove $n - (2^k - 1)$ stones. This is at least 1, and at most $\frac{n}{2}$ because $n \leq 2^{k+1} - 2$. So the guess is right by Theorem 1.
Since $1023 < 2026 < 2047$, the first player wins by removing $2026 - 1023 = 1003$ stones (allowed, as $1003 \leq 1013$), and thereafter always moves to the next smaller number of the form $2^k - 1$. The moves from 2026 reach $1013, \ldots, 2025$, and 1023 is the only P-position among them, so this is the only winning first move.

1.2 Symmetry and pairing

The opening game used a symmetry strategy. One player makes the position symmetric and then copies. A pairing strategy is the same idea without geometry. Split the possible moves into pairs

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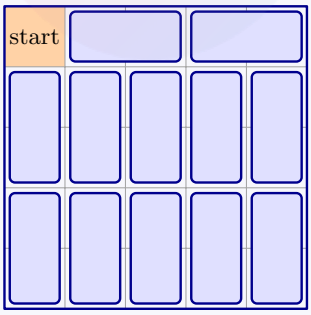
so that whenever the opponent takes one element of a pair, the other is still available and is a good reply. In both cases the proof has two parts, and the second is the one students forget.

1. The reply is always legal (the mirror point or partner is still free, and the move is allowed).
2. The reply does not itself do damage (the reply does not complete a losing pattern and does not interact with the opponent's last move).

Point (2) fails when an object is its own mirror image, for instance a domino covering the centre of a board. The usual fix is for the first player to occupy the self-symmetric object first, as in the chord game.

Example 2.
Alice places a token on any cell of an $m \times n$ board. Then Bob and Alice take turns, Bob first, to move the token to a cell sharing a side with its current cell that the token has never visited. The player who cannot move loses. For which m, n does Bob win?

Solution.
Small cases. On a 1×2 board Bob wins, since he moves to the other cell. On a 1×3 board Alice starts at an end, Bob moves to the middle and Alice to the other end. Two cells adjacent to each other suggest a domino, so try a pairing strategy based on a tiling by dominoes.
If mn is even, some side is even and the board can be tiled by dominoes. Bob always moves the token to the other cell of the domino containing the token. We claim that after each of Bob's moves the visited cells form a union of whole dominoes. This holds after his first move. If the claim holds and Alice moves to an unvisited cell, that cell's domino is untouched, so its partner is unvisited and adjacent, and Bob's move restores the claim. Hence Bob can always move, and since the game is finite, Alice loses.
If mn is odd, Alice starts in a corner. The rest of the board can be tiled by dominoes, namely the remaining $n - 1$ cells of the first row by horizontal dominoes, and the other $m - 1$ rows, an even number, by vertical ones. Now Alice plays Bob's strategy from the even case. The visited cells are always the corner together with whole dominoes after each of her moves, so she can always reply. Bob wins exactly when mn is even.



The same proof shows more, and is worth remembering. In this game on any graph, Bob wins if and only if the graph has a perfect matching. If there is none, Alice takes a maximum matching M, starts at a vertex v not covered by M, and always moves along M. If Bob ever reached a vertex not covered by M, the path from v would alternate between edges outside and inside M and join two uncovered vertices. Swapping the edges along that path would give a larger matching, which is impossible. (The method of augmenting paths.)

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1.3 Strategy stealing

Sometimes we can prove that the first player wins without finding a strategy. The classical example is Chomp, played on an $m \times n$ chocolate bar, not 1×1, whose bottom-left square is poisoned. A move is to choose a remaining square and eat that square together with every square that is both weakly above and weakly to the right of the chosen square. Whoever eats the poisoned square loses.

Theorem 2.
In Chomp the first player has a winning strategy.

Proof. The game is finite with no draws, so by Theorem 1 one of the players has a winning strategy. Suppose the second player has the winning strategy. Then in particular the second player has a winning reply R to the first move "eat the top-right square". But every move eats the top-right square, so the position after the first move and R is the position after the single move R. The first player could therefore play R at once and then follow the second player's strategy, winning. This contradicts the assumption. ■

The argument needs a first move that never hurts. Here, every position reachable after that move and a reply is reachable by a single move. In achievement games (first to complete a pattern wins) the analogue is that owning an extra square never hurts, and stealing shows that the second player has no winning strategy. Strategy stealing does not tell you the strategy, and nobody knows one for general rectangular Chomp.

1.4 Processes and termination

From L4-CO1. A process stops if a non-negative integer decreases at every step. Two extensions are often needed at this level.

1. If there are finitely many states, any real-valued quantity that changes in one direction at every step proves termination, since no state can repeat.
2. Compare vectors lexicographically. If each step decreases $(v_1, \ldots, v_k)$ in lexicographic order and each v_i is a non-negative integer bounded in terms of the earlier coordinates, the process stops.

When asked where a process ends, look at what a final state must satisfy, since "no move is possible" is usually a strong condition.

Example 3.
Each vertex of a graph is coloured red or blue. A move. Choose a vertex more than half of whose neighbours have its own colour, and switch that colour. Prove that the moves must stop. Does this remain true if "more than half" is replaced by "at least half"?

Solution.
Let M be the number of edges whose ends have the same colour. Suppose the chosen vertex v has s neighbours of its own colour and t of the other, with $s > t$. Switching v changes only the edges at v. The s monochromatic ones become bichromatic and the t bichromatic ones become monochromatic. So M decreases by $s - t \geq 1$. Since M is a non-negative integer, the moves stop after at most e steps.
With "at least half" the same computation gives a decrease of $s - t \geq 0$, which proves

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nothing, and the claim is false. Take a path a, v, b with a red, b blue and v red. Then v has exactly half of its neighbours of its own colour, so v may switch to blue. Then again exactly half, so v may switch back, for ever.
The process also gives an existence result. In a final state no vertex has more than half its neighbours of its own colour. Since a final state exists, every graph has a red/blue colouring in which each vertex has at least half its neighbours of the other colour. (A process with a monovariant gives the same conclusion as choosing a colouring with M minimal. The two arguments are the same.)

1.5 Constructions and the averaging method

In "find the largest" problems the answer is a bound together with an example meeting the bound, and the equality case of the bound tells you what the example must look like (as in Example 2 of L5-CO1). When an explicit construction is hard, the averaging method proves existence instead. If a quantity has average μ over a finite collection of objects, some object has the quantity at least μ, and some has the quantity at most μ. Phrased with probability, if a random object has expected value μ, then some object attains a value at least μ.

Theorem 3 (Erdős).
If $\binom{n}{k}2^{1-\binom{k}{2}} < 1$, the edges of K_n can be coloured red and blue with no monochromatic K_k. In particular $R(k,k) > 2^{k/2}$ for $k \geq 3$.

Proof. Among all $2^{\binom{n}{2}}$ colourings, a fixed set of k vertices is monochromatic in $2 \cdot 2^{\binom{n}{2}-\binom{k}{2}}$ of them. Adding over the $\binom{n}{k}$ sets, the number of colourings with some monochromatic K_k is at most $\binom{n}{k}2^{1-\binom{k}{2}} \cdot 2^{\binom{n}{2}}$, less than the total. For $n = \lfloor 2^{k/2} \rfloor$ we have $\binom{n}{k} < \frac{n^k}{k!} \leq \frac{2^{k^2/2}}{k!}$, so $\binom{n}{k}2^{1-\binom{k}{2}} < \frac{2^{1+k/2}}{k!} < 1$ for $k \geq 3$. ■

No explicit colouring anywhere near this good is known. The next theorem uses a random ordering instead of a random colouring.

Theorem 4 (Caro, Wei).
Every graph has an independent set of size at least $\sum_v \frac{1}{d(v)+1}$.

Proof. Order the vertices uniformly at random and let S be the set of vertices that come before all of their neighbours. Two adjacent vertices cannot both come before each other, so S is independent. A vertex v is in S exactly when v is first among the $d(v)+1$ vertices of $\{v\} \cup N(v)$, which has probability $\frac{1}{d(v)+1}$. So the expected size of S is $\sum_v \frac{1}{d(v)+1}$, and some ordering gives an S at least that large. ■

Example 4.
A club has 30 members and 60 pairs of friends. Find the largest k such that one can always find k members no two of whom are friends.

Solution.
First guess the extremal example. To keep every set of pairwise strangers small, the friendships should be packed into cliques, so try disjoint cliques of equal size, namely six groups of five, all friends within a group. That uses $6\binom{5}{2} = 60$ friendships, and a set of pairwise strangers has at most one member from each group. So $k \leq 6$.

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For the bound, Theorem 4 gives an independent set of size at least $\sum_v \frac{1}{d(v)+1}$. Here $\sum_v d(v) = 120$ by the handshake lemma, so $\sum_v (d(v)+1) = 150$, and by the Cauchy-Schwarz (or AM-HM) inequality

$$\sum_v \frac{1}{d(v)+1} \geq \frac{30^2}{\sum_v (d(v)+1)} = \frac{900}{150} = 6.$$

So 6 pairwise strangers can always be found, and the answer is $k = 6$.
Note that the bound and the example match because equality in Cauchy-Schwarz needs all degrees equal, and equality in Theorem 4 holds for disjoint cliques. In general Theorem 4 implies Turán's theorem in its complementary form.

Common traps.

- Describing a strategy without proving that the reply is always legal. Say why the mirror move or partner is still available.
- A symmetry strategy that breaks on a self-symmetric move. Look for objects fixed by the symmetry.
- Using Theorem 1 for a game that can go on for ever, or that has draws. Check finiteness first.
- Expecting strategy stealing to produce a strategy. Strategy stealing proves existence only.
- A monovariant that changes by "at least 0". This proves nothing, as Example 3 shows.
- In an averaging argument, averaging over a set in which not every object is allowed (for instance, splits into two groups when the problem asks for equal groups).

1.6 How to spot this in a contest

For a game, in this order:

1. play small cases and tabulate P-positions,
2. look for a symmetry or a pairing of moves, and check the self-symmetric moves,
3. if the first player should win but no strategy appears, try strategy stealing,
4. for "can Breaker stop Maker" questions, try a pairing, and if that is impossible, a weight function (practice problem 6).

For a process, ask what one general move does to sums, sums of squares, numbers of monochromatic pairs, and the lexicographic order, and separately what a final state must look like. For "find the largest" problems, never stop after the bound. Write down the example, and if the example is hard to find, see whether averaging over a natural family of candidates (random colourings, random orderings, random splits) proves that a good one exists.

2 Practice problems

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The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ Alice and Bob take turns to colour cells of an $n \times n$ board, where $n \geq 4$. Alice colours an uncoloured cell red, then Bob colours an uncoloured cell blue, and so on until every cell is coloured. Alice wins if some row or some column is entirely red. Prove that Bob can prevent this.

Problem 2 ⋆⋆ A pile contains 100 stones. Two players alternately remove stones. In each move a player must remove at least one stone and at most one third of the stones currently in the pile. The player who cannot move loses. Which player has a winning strategy, and what are the winning first moves?

Problem 3 ⋆⋆⋆ Alice and Bob take turns to claim numbers from $\{1, 2, \ldots, n\}$ not yet claimed, Alice first. The first player to have claimed three different numbers, one of which is the sum of the other two, wins. If all numbers are claimed and neither player has done this, the game is a draw.

(a) Prove that, for every n, Bob has no winning strategy.

(b) Prove that Alice has a winning strategy if $n \geq 7$.

Problem 4 ⋆⋆⋆ There are $2n$ points on a circle, joined in pairs by n chords, so that each point is an end of exactly one chord. A move. Choose two chords AB and CD that cross, and replace them by the chords AC and BD, or by the chords AD and BC, choosing either option. Prove that, however the moves are chosen, after at most $\frac{n(n-1)}{2}$ moves no two chords cross.

Problem 5 ⋆⋆⋆ A school has $2k$ pupils ($k \geq 1$), and e pairs of them are friends. Prove that the pupils can be split into two classes of k pupils each so that at least $\frac{ek}{2k-1}$ pairs of friends are in different classes. Show that for every k there is a set of friendships for which this bound cannot be improved.

Problem 6 ⋆⋆⋆⋆ A family $\mathcal{F}$ of m sets is given, each consisting of exactly k elements of a finite set X, where $m < 3^{k-1}$. Alice and Bob take turns to claim elements of X not yet claimed, Alice first, until every element is claimed. On each of her turns Alice claims one element, and on each of his turns Bob claims two elements (or one, if only one is left). Alice wins if she claims every element of some set in $\mathcal{F}$. Prove that Bob can prevent this. (For example, Bob can stop Alice when $\mathcal{F}$ consists of 200 sets of 6 elements each.)

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **IMO 1986 Problem 3** ⋆⋆
   Look for a quantity that every move changes in the same direction.

2. **BMO2 2016/17 Problem 4** ⋆⋆⋆
   You need a strategy and also an argument that no strategy with fewer attempts can always succeed.

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3. **USAMO 1999 Problem 5** ⋆⋆⋆
   Look for positions in which every move loses, and think about parity.

4. **BMO2 2017/18 Problem 2** ⋆⋆⋆
   Try small n. For the cases where the Hatter fails, find a set of seats Alice can keep avoiding.

5. **USAMO 2004 Problem 4** ⋆⋆⋆
   Decide first which player should win, then look for a pairing of cells.

6. **BMO2 2023/24 Problem 4** ⋆⋆⋆
   Find the necessary conditions from invariants first, then build a sequence of moves for the remaining cases.

7. **IMO 2022 Problem 1** ⋆⋆⋆
   Experiment with small n and track how the number of chains changes with each move.

8. **IMO 2025 Problem 5** ⋆⋆⋆
   Try simple strategies for each player first, and see which of the two constraints each one strains.

9. **USAJMO 2014 Problem 5** ⋆⋆⋆⋆
   For B's side, find a pattern of cells B can keep from filling that meets every long enough line.

10. **USAMO 2010 Problem 6** ⋆⋆⋆⋆
    One direction is an averaging argument over random choices. The other needs a well-chosen board.

11. **IMO 2013 Problem 2** ⋆⋆⋆⋆
    Treat the bound and the construction separately. For the lower bound, look for a configuration of points that is hard to separate.

12. **IMO 2024 Problem 5** ⋆⋆⋆⋆
    Work out what one failed attempt can tell Turbo, and argue from both sides.

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4 Hints

1. Find $2n$ disjoint pairs of cells such that each row and each column contains one of the pairs entirely.
2. From n stones you can move to exactly the numbers from $\lceil 2n/3 \rceil$ to $n-1$. Once you know one P-position q, the next one is the smallest m from which q cannot be reached.
3. (a) Steal a strategy. An extra number never hurts Alice. (b) Claim 1 first. Which pairs of numbers then complete a triple with 1?
4. Count the pairs of chords that cross. For another chord ℓ, how many of A, B, C, D can lie on each side of ℓ, and how many of the two old and the two new chords does ℓ cross in each case?
5. Average over all splits into two classes of k. In what fraction of them is a given pair of friends separated?
6. Give each set that contains none of Bob's elements a weight, multiplied by a fixed factor whenever Alice claims one of its elements, and let Bob keep the total weight small. Which factor matches two elements per turn for Bob?

5 Solutions

Problem 1

Alice and Bob take turns to colour cells of an $n \times n$ board, where $n \geq 4$. Alice colours an uncoloured cell red, then Bob colours an uncoloured cell blue, and so on until every cell is coloured. Alice wins if some row or some column is entirely red. Prove that Bob can prevent this.

Number the rows and columns $0, 1, \ldots, n-1$ and read indices modulo n. For row i take the pair $\{(i,i),(i,i+1)\}$. These $2n$ cells contain exactly two cells of each column j, namely (j,j) and $(j-1,j)$. Since $n \geq 4$, column j has at least two further cells. Choose two of them as the pair for column j. The $2n$ pairs are disjoint. Row pairs lie in different rows, column pairs lie in different columns, and no cell of a column pair is in a row pair.

Bob's strategy. Whenever Alice colours a cell belonging to a pair whose other cell is uncoloured, Bob colours that other cell. Otherwise he colours any uncoloured cell. We claim that Alice never owns both cells of a pair. Suppose Alice colours a cell x of a pair $\{x,y\}$. If y is uncoloured, Bob takes y at once. If y is already coloured, y is not red. Had Alice coloured y earlier, Bob would have answered by colouring x, and x would not now be available.

Every row and every column contains a pair, and no pair is entirely red. So no row or column is entirely red.

(For $n = 2$ Alice wins. After her first cell, Bob cannot block both the row and the column through that cell. For $n = 3$ no pairing exists, since $2n = 6$ disjoint pairs would need $12 > 9$ cells. Bob still wins for $n = 3$, but the only proof I know is a computer search of the game tree.)

Problem 2

A pile contains 100 stones. Two players alternately remove stones. In each move a player must remove at least one stone and at most one third of the stones currently in the pile. The player who cannot move loses. Which player has a winning strategy, and what are the winning first moves?

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Removing j stones with $1 \le j \le \lfloor n/3 \rfloor$ from a pile of n leaves m stones with $n - \lfloor n/3 \rfloor \le m \le n-1$, that is, $\lceil 2n/3 \rceil \le m \le n-1$, and every such m can be reached. Write $g(n) = \lceil 2n/3 \rceil$. Note that g is non-decreasing. Piles of 1 and 2 stones allow no move.

Define $q_1 = 1$ and, for $i \ge 1$, let q_{i+1} be the least m with $g(m) > q_i$, that is, the least m with $\frac{2m}{3} > q_i$, so $q_{i+1} = \lfloor \frac{3q_i}{2} \rfloor + 1$. This gives

$$1,\ 2,\ 4,\ 7,\ 11,\ 17,\ 26,\ 40,\ 61,\ 92,\ 139,\ \ldots$$

We prove that the P-positions are exactly the q_i, by checking the two properties of Theorem 1. If $q_i < n < q_{i+1}$, then $g(n) \le q_i$ by the definition of q_{i+1}, so q_i can be reached from n. From $n = q_{i+1}$, the reachable numbers lie between $g(q_{i+1}) > q_i$ and $q_{i+1} - 1$, so no q_j can be reached. Finally $q_1 = 1$ has no moves. (Formally this is an induction on n.)

Since $92 < 100 < 139$, the pile of 100 is an N-position and the first player wins. The moves from 100 reach $67, 68, \ldots, 99$, and the only q_i in this range is 92. So the only winning first move is to remove 8 stones, and afterwards the first player always moves to the next smaller q_i.

Answer: The first player wins; the only winning first move is to remove 8 stones.

Problem 3

Alice and Bob take turns to claim numbers from $\{1, 2, \ldots, n\}$ not yet claimed, Alice first. The first player to have claimed three different numbers, one of which is the sum of the other two, wins. If all numbers are claimed and neither player has done this, the game is a draw.

(a) Prove that, for every n, Bob has no winning strategy.

(b) Prove that Alice has a winning strategy if $n \ge 7$.

Call three different numbers, one of which is the sum of the other two, a *triple*.

(a) Suppose Bob has a winning strategy σ. Alice plays as follows. She first claims any number e, her *extra* number. From then on she imagines a game in which Bob is the first player and she is the second player using σ, and in which her extra number is unclaimed. Bob's real moves are his moves in the imagined game. When σ tells her to claim a number x, she claims x if x is unclaimed. If x is her extra number, she already owns that number. She claims any unclaimed number instead (if there is none, the game is over) and that number becomes her new extra. By induction on the number of moves, at every moment Bob owns the same numbers in both games, and Alice owns in the real game every number she owns in the imagined one.

Since σ wins, at some moment of the imagined game Alice completes a triple, and Bob has not completed one before. At the corresponding moment of the real game Bob still has not completed a triple, while Alice's numbers contain her imagined numbers, so she has completed a triple by then. Hence this strategy wins for Alice against every play of Bob. But Alice and Bob cannot both have winning strategies (play them against each other). This contradiction shows that Bob has no winning strategy.

(b) Let $n \ge 7$. Alice claims 1, and Bob claims some number y. The triples containing 1 include $\{1, z-1, z\}$ and $\{1, z, z+1\}$ for $3 \le z \le 6$. The four sets $\{2,3,4\}$, $\{3,4,5\}$, $\{4,5,6\}$, $\{5,6,7\}$ have no common element, so one of them, say $\{z-1, z, z+1\}$, does not contain y. Alice claims z. Now each of $z-1$ and $z+1$ is unclaimed, and claiming either completes a triple with 1 and z. Bob has only two numbers after his next move, so he cannot complete a triple at that point, and he can claim only one of $z-1$, $z+1$. Alice claims the other one on her third move and wins.

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(A computer search shows that for $n \leq 6$ the game is a draw, so the bound in (b) cannot be lowered.)

Problem 4

There are $2n$ points on a circle, joined in pairs by n chords, so that each point is an end of exactly one chord. A move. Choose two chords AB and CD that cross, and replace them by the chords AC and BD, or by the chords AD and BC, choosing either option. Prove that, however the moves are chosen, after at most $\frac{n(n-1)}{2}$ moves no two chords cross.

Two chords with four different ends cross exactly when the ends of one lie on different arcs cut off by the other. Let X be the number of pairs of chords that cross. We prove that every move decreases X by at least 1. Since $X \leq \binom{n}{2}$ at the start and $X \geq 0$, at most $\binom{n}{2}$ moves can be made, and when no move is possible, no two chords cross.

Consider a move on the crossing chords AB and CD. As they cross, the points lie on the circle in the cyclic order A, C, B, D. The two new chords do not cross each other. The ends of AC are consecutive in this order, and so are those of BD, and the same holds for AD and BC. Pairs of chords not involving A, B, C, D are unchanged. So we only need to show that every other chord ℓ crosses at most as many of the two new chords as of the two old ones.

The chord ℓ has no end among A, B, C, D, and ℓ cuts the circle into two arcs. A chord with ends among A, B, C, D crosses ℓ exactly when its ends are on different arcs. If all four points are on one arc, ℓ crosses none of the chords. If one arc contains exactly one of the four points, then ℓ crosses exactly one chord of each pair, namely the one through that point. If each arc contains two of the points, these two are consecutive in the cyclic order A, C, B, D, so they are $\{A, C\}$ and $\{B, D\}$, or $\{C, B\}$ and $\{D, A\}$. In both cases A and B are on different arcs, and so are C and D, so ℓ crosses both old chords, and ℓ crosses at most two new chords. This proves the claim, so X drops by at least 1, the crossing of AB and CD itself.

(The total length of the chords also decreases at each move, because the diagonals of the convex quadrilateral $ACBD$ are together longer than either pair of opposite sides. This proves that the process stops, but gives no bound on the number of moves.)

Problem 5

A school has $2k$ pupils ($k \geq 1$), and e pairs of them are friends. Prove that the pupils can be split into two classes of k pupils each so that at least $\frac{ek}{2k-1}$ pairs of friends are in different classes. Show that for every k there is a set of friendships for which this bound cannot be improved.

Consider all $\binom{2k}{k}$ ways to choose the pupils of the first class. Each split into two classes of k arises from exactly two such choices, which does not affect averages. For a fixed pair of friends $\{x, y\}$, the choices in which x and y are in different classes are those containing exactly one of them, giving $2\binom{2k-2}{k-1}$ choices. Since

$$\frac{2\binom{2k-2}{k-1}}{\binom{2k}{k}} = \frac{2 \cdot k \cdot k}{2k(2k-1)} = \frac{k}{2k-1},$$

each pair of friends is separated in a fraction $\frac{k}{2k-1}$ of the choices. Adding over the e pairs, the average number of separated pairs over all choices is $\frac{ek}{2k-1}$, so some choice separates at least $\frac{ek}{2k-1}$ pairs.

If every two pupils are friends, then $e = \binom{2k}{2} = k(2k-1)$ and the bound is k^2. Every split separates exactly $k \cdot k = k^2$ pairs, so the bound cannot be improved in this case.

Kevinvariant
kevinvariant08.github.io

Problem 6

A family $\mathcal{F}$ of m sets is given, each consisting of exactly k elements of a finite set X, where $m < 3^{k-1}$. Alice and Bob take turns to claim elements of X not yet claimed, Alice first, until every element is claimed. On each of her turns Alice claims one element, and on each of his turns Bob claims two elements (or one, if only one is left). Alice wins if she claims every element of some set in $\mathcal{F}$. Prove that Bob can prevent this. (For example, Bob can stop Alice when $\mathcal{F}$ consists of 200 sets of 6 elements each.)

Call a set of $\mathcal{F}$ *alive* if Bob has claimed none of its elements, give an alive set A the weight $w(A) = 3^{a(A)}$, where $a(A)$ is the number of elements of A claimed by Alice, and let Φ be the sum of the weights of the alive sets. At any moment, for an unclaimed element z let $\varphi(z)$ be the sum of the weights of the alive sets containing z.

If Alice ever claims all of some $A \in \mathcal{F}$, then A is alive (Bob owns none of its elements) and $w(A) = 3^k$, so $\Phi \geq 3^k$ at that moment. We give a strategy for Bob that keeps $\Phi < 3^k$ after each of Alice's moves. This proves that Alice cannot win.

Bob's strategy. On each turn, claim an unclaimed element y_1 with $\varphi(y_1)$ as large as possible, and then, with φ recomputed after this claim, an unclaimed element y_2 with $\varphi(y_2)$ as large as possible.

At the start $\Phi = m$. Alice's first move triples the weights of the sets containing her element and leaves the others alone, so afterwards $\Phi \leq 3m < 3^k$.

Now consider a turn of Bob followed by a move x of Alice. Since an element was left for Alice, Bob claimed two elements y_1, y_2 on that turn. Write φ_0, φ_1 and φ_2 for the values of φ before Bob's turn, after he claims y_1, and after he claims y_2. Claiming y_1 makes the alive sets containing y_1 dead, so Φ decreases by $\varphi_0(y_1)$. Similarly claiming y_2 decreases Φ by $\varphi_1(y_2)$. Bob's claims change no weights and only remove sets, so $\varphi_2(x) \leq \varphi_1(x) \leq \varphi_0(x)$. Alice's move triples the weight of every alive set containing x, so Φ increases by $2\varphi_2(x)$. Since x was unclaimed when Bob chose y_1 and y_2, his choices give $\varphi_0(x) \leq \varphi_0(y_1)$ and $\varphi_1(x) \leq \varphi_1(y_2)$. Hence

$$2\varphi_2(x) \;\leq\; \varphi_0(x) + \varphi_1(x) \;\leq\; \varphi_0(y_1) + \varphi_1(y_2),$$

so Bob's turn and Alice's next move together do not increase Φ.

So $\Phi < 3^k$ after every move of Alice, and Alice never claims a whole set of $\mathcal{F}$. In the example, $200 < 3^5 = 243$.

(With weights $2^{a(A)}$ and one element per turn for Bob, the same argument shows that Bob can stop Alice whenever $m < 2^{k-1}$. This is a theorem of Erdős and Selfridge. In the example this bound is no use, since $200 > 2^5$. The bound 3^{k-1} is sharp for $k = 2$, as the three sets $\{1,2\}$, $\{1,3\}$, $\{1,4\}$, Alice claims 1, Bob can claim only two of 2, 3, 4, and Alice claims the third.)